
A P P R O X I M A T E B A Y E S I A N C O M P U T A T I O N F O R I D E N T I T Y B Y D E S C E N T A N A L Y S I S

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From 1974 to 1983, Argentina was ruled by a military dictatorship.



It is estimated that 22,000 to 30,000 people
“disappeared” or were killed, many abducted without any
public record of arrest or trial.

Victims included students, trade unionists, intellectuals, journalists, activists, and ordinary citizens suspected of political dissent.

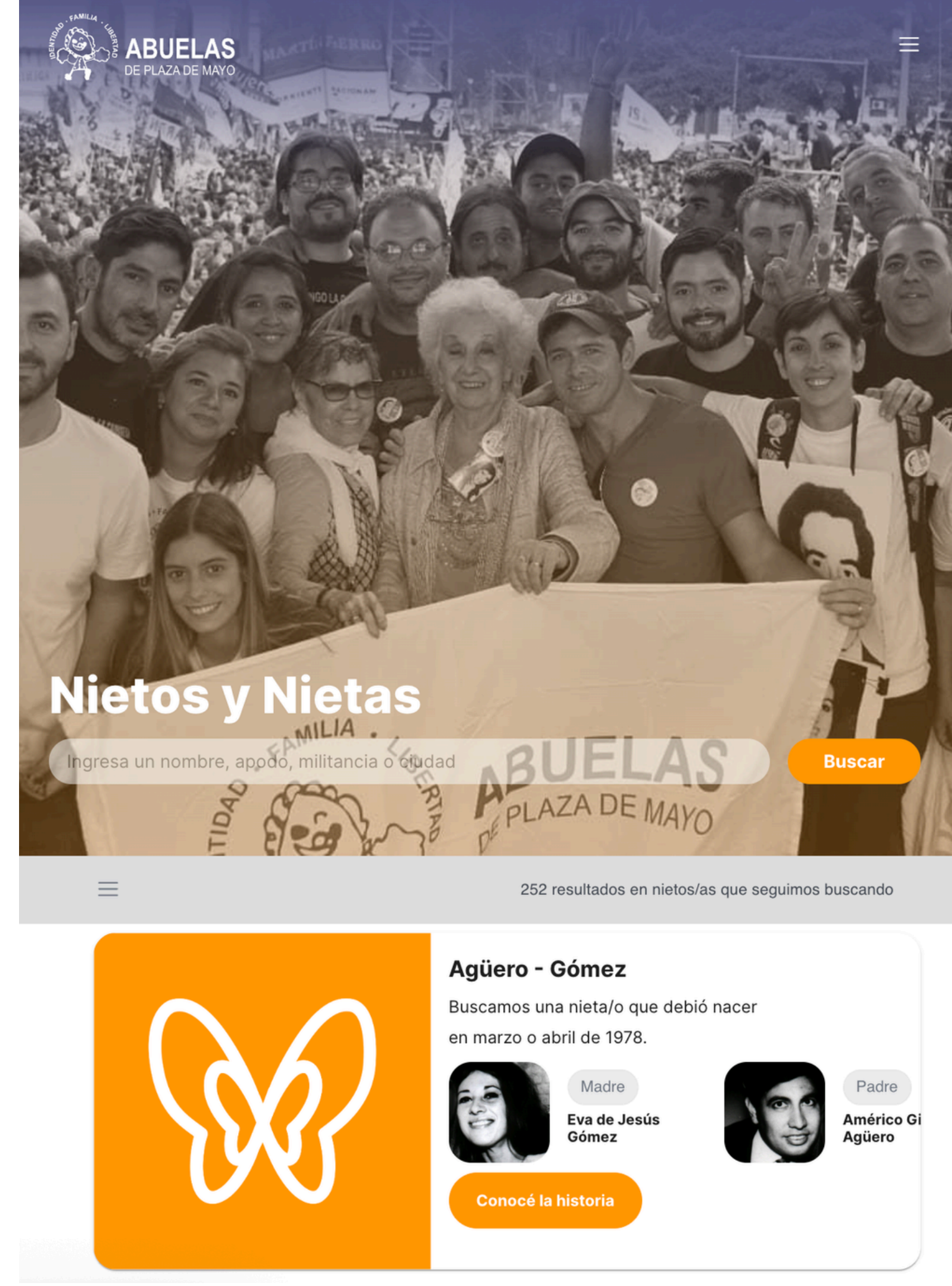
Common practices included torture, clandestine detention centers, and forced “disappearances” (los desaparecidos) where victims were taken away in secret and often never heard from again.

Some children born to detained mothers were taken from their biological families and given to other families, sometimes without any record of their origin.



Human rights groups such as the Abuelas de Plaza de Mayo (Grandmothers of Plaza de Mayo) have been central in investigating and advocating for the recovery of these children.

There are currently 252 open cases submitted to Abuelas de Plaza Mayo.



The screenshot shows the website of Abuelas de Plaza de Mayo. At the top, there is a header with the organization's logo and name. Below the header is a large image of a group of people, including a woman in the center. Overlaid on this image is the text 'Nietos y Nietas' and a search bar with the placeholder text 'Ingresa un nombre, apodo, militancia o ciudad'. To the right of the search bar is an orange button labeled 'Buscar'. Below the search bar, a grey bar indicates '252 resultados en nietos/as que seguimos buscando'. The main content area features a large orange square with a white butterfly logo on the left. To the right of the logo, the text 'Agüero - Gómez' is displayed, followed by the description 'Buscamos una nieta/o que debió nacer en marzo o abril de 1978.' Below this text are two small circular portraits: one of a woman labeled 'Madre' and 'Eva de Jesús Gómez', and one of a man labeled 'Padre' and 'Américo Gi Agüero'. At the bottom of this section is an orange button labeled 'Conocé la historia'.

ABUELAS DE PLAZA DE MAYO

Nietos y Nietas

Ingresa un nombre, apodo, militancia o ciudad

Buscar

252 resultados en nietos/as que seguimos buscando

Agüero - Gómez

Buscamos una nieta/o que debió nacer en marzo o abril de 1978.

Madre
Eva de Jesús Gómez

Padre
Américo Gi Agüero

Conocé la historia

CURRENT METHODS

Pairwise LR (Likelihood Ratio method)

$$LR = \frac{P(\text{observed data} \mid H_1)}{P(\text{observed data} \mid H_2)}$$

Compares two individuals at a time using their shared IBD segments. Calculates how likely the observed sharing is under different relationship hypotheses (e.g., sibling vs. cousin). Works well for close relatives but struggles with complex or distant pedigrees since it ignores information from other family members.

Gives the relative likelihood.

CREST (multi-way IBD-based method)

Developed to improve upon pairwise inference. It still focuses on a target pair of individuals, but it uses information from mutual relatives who share IBD with both individuals. This is called triangulation.

By considering how DNA segments are co-shared among multiple people, CREST improves classification accuracy (especially for 2nd-4th degree relatives).

Deterministic classifier, so it does not quantify uncertainty.

It is probable that the children who went missing when they were babies or before they were born grew up without knowing their true background, and now have families of their own.

We are looking for a granddaughter/son who must have been born in May or June 1975.

I doubt my origin

I want to contribute information

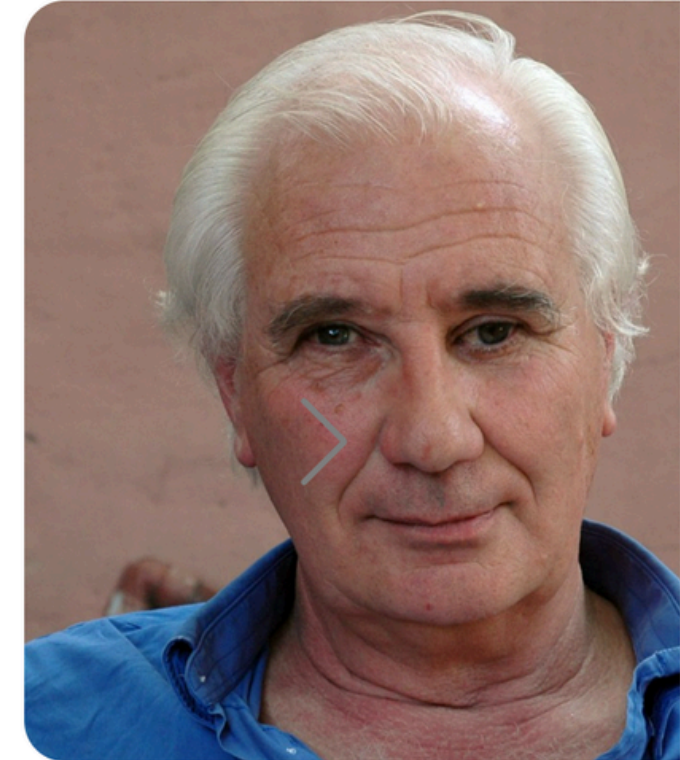
The Orzaocoa-Gómez family



Mother

Maria De Las Mercedes Gomez

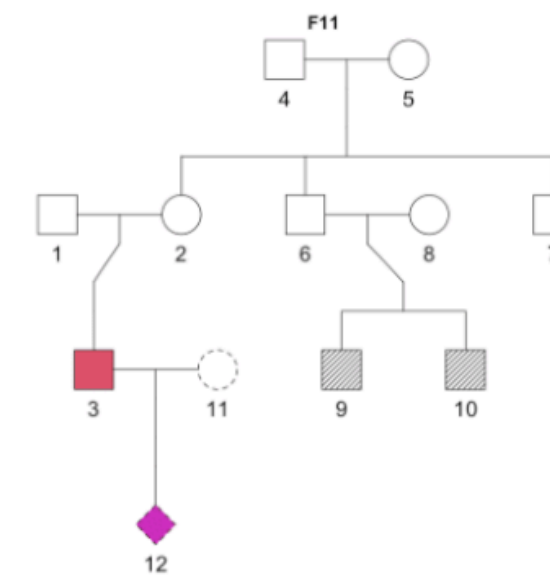
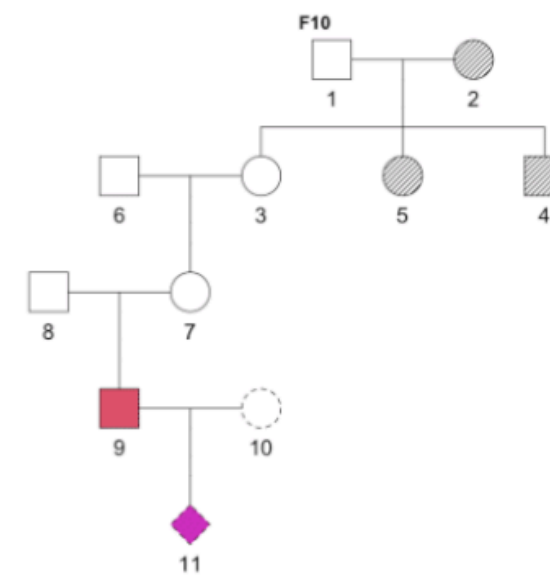
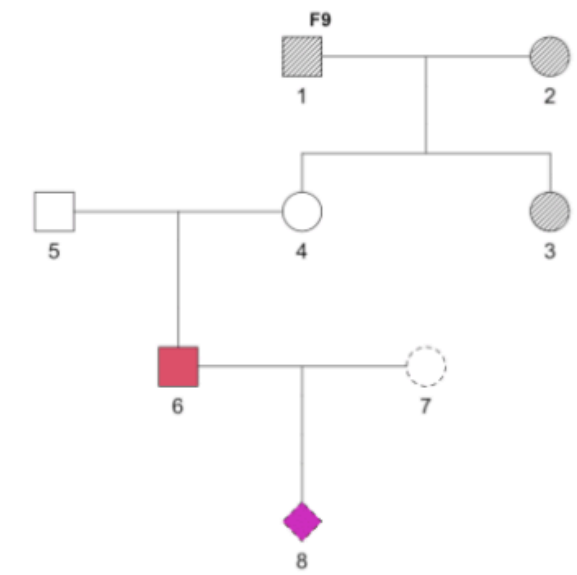
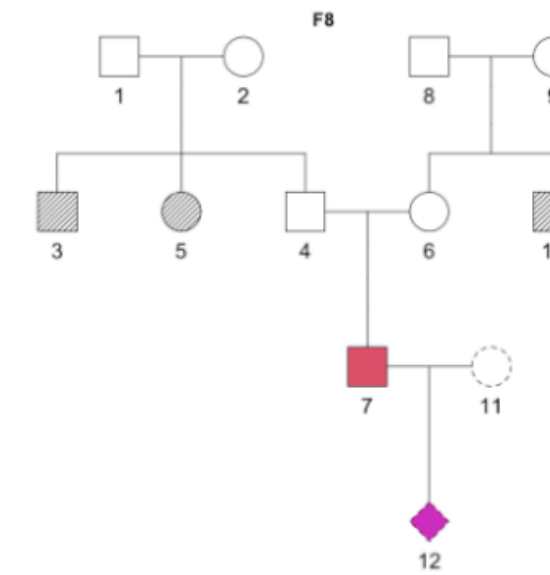
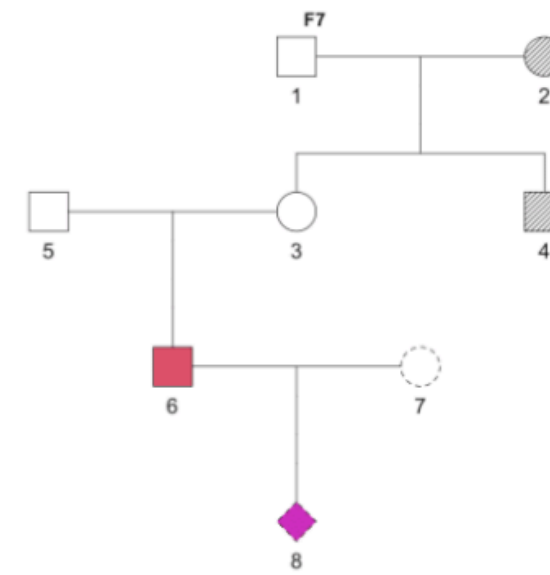
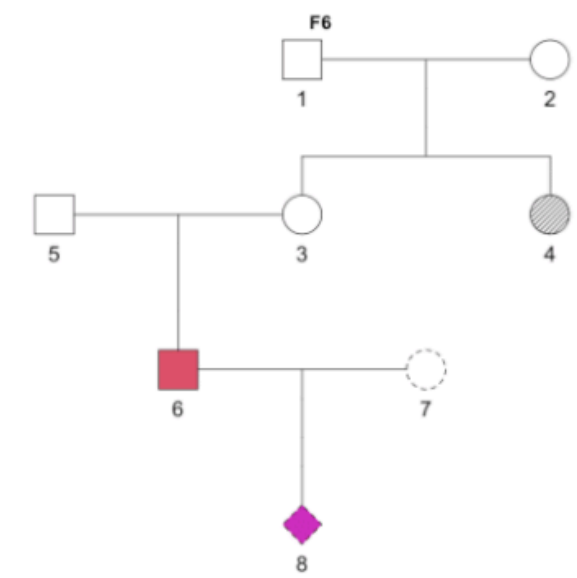
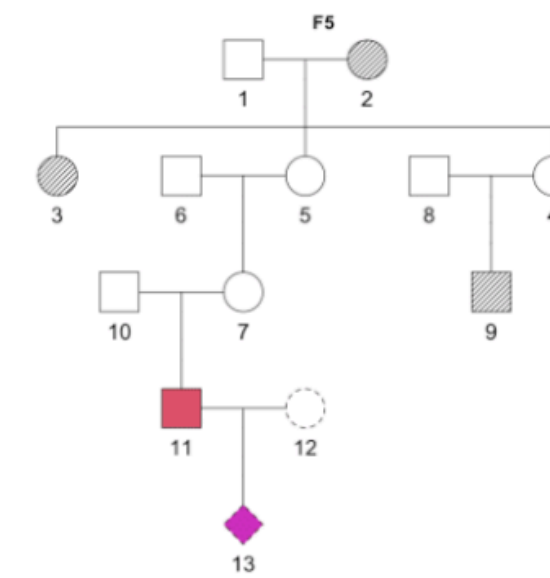
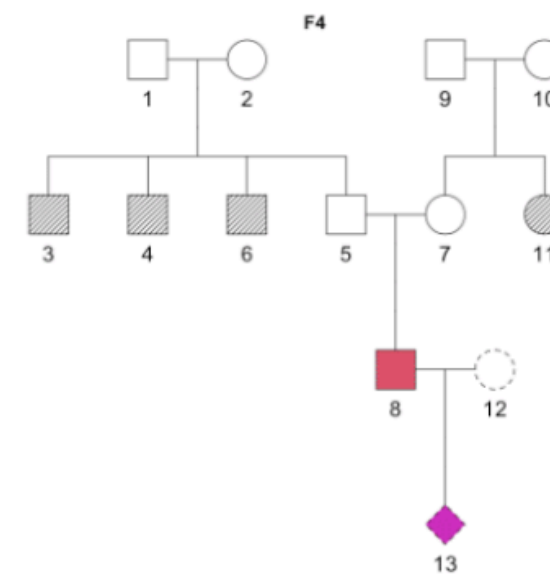
Disappeared on March 21, 1975, in the city of Córdoba. Six or seven months pregnant.



Father

Carlos Normando Orzaocoa

From Abuelas, we continue searching for your daughter or son.



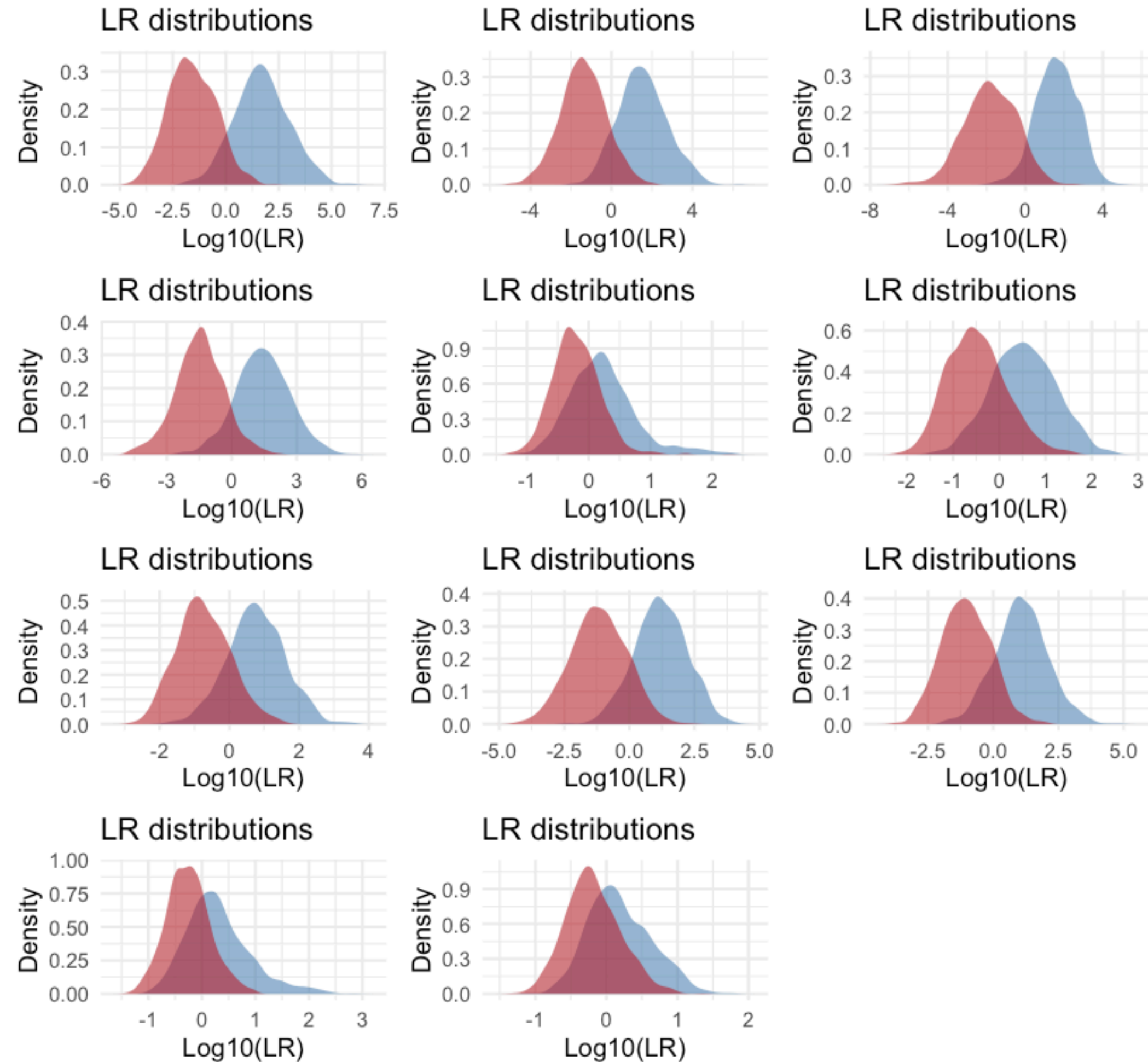
Using 10 cases from “Making decisions in missing person identification cases with low statistical power” by Marsico et al. (FSIG, 2021), along with one case from Abuelas de Plaza de Mayo, and extending each pedigree by one generation.

Pairwise LR method

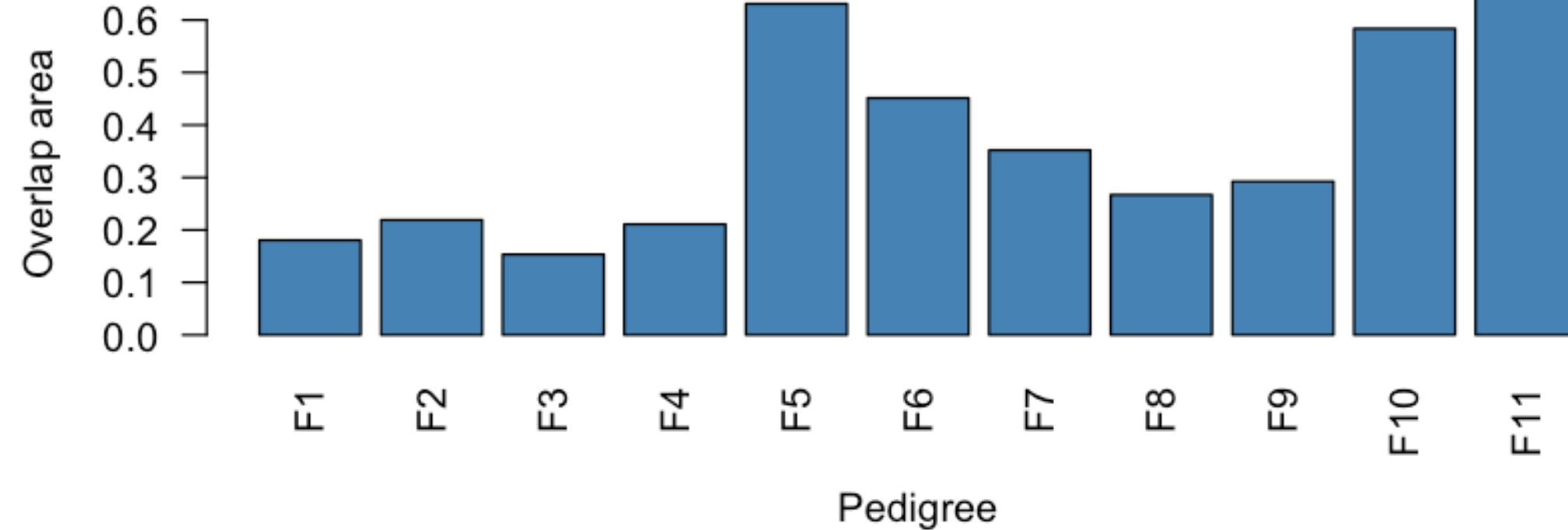
With rounds of 10,000 simulations per hypothesis (H1: Unidentified Person is Missing Person and H2: UP is not MP) I obtained the results presented in the plot.

LR distributions (H1 vs H2) for all 11 pedigrees

Red = unrelated; Blue = true-child

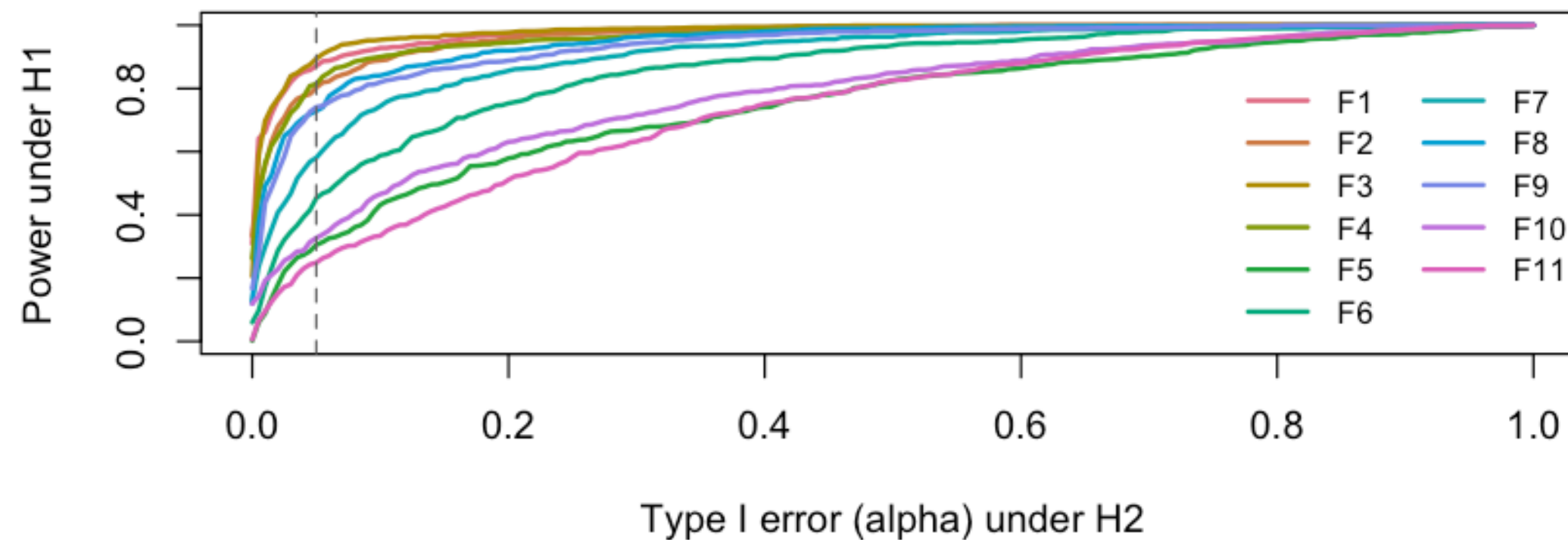


H1 vs H2 density overlap by pedigree



Curves that rise quickly toward 1 mean LR separates H1 from H2 well for that pedigree.

Power curves from LR thresholding



Curves that stay low until large alpha mean you can only get decent true-positive rates if you tolerate lots of false positives.

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Goal

Estimate posterior probabilities of competing hypotheses when the likelihood is too complex or unknown.

Approach

- a. Simulate data under each hypothesis
- b. Compute summary statistics for each simulated dataset.
- c. Compare simulated and observed summaries using a distance metric.
- d. Accept simulations that are “close enough” (within tolerance ϵ).
- e. Approximate the posterior from accepted simulations.

Output

Relative posterior probabilities for each relationship hypothesis.

Strengths

Works with complex pedigrees.

Naturally incorporates prior beliefs and uncertainty.

Can implement multi-way analysis into the summary statistics

Limitations

Requires informative summary statistics.

Computationally intensive for many hypotheses.

Summary statistics I used (per query-known pair)

For each pair, I compute 9 segment-based summaries:

1. Total IBD (sum of segment lengths)
2. Number of segments
3. Largest segment
4. Proportion of segments > 10 cM
5. Proportion of segments > 20 cM
6. Mean segment length
7. Median segment length
8. Largest / Total IBD ratio
9. Chromosomes shared (count of chromosomes with any segment)

This captures not only “how much IBD” but “how it is shared”.

How I handled multiple typed individuals ($m > 1$)

Each pedigree has:

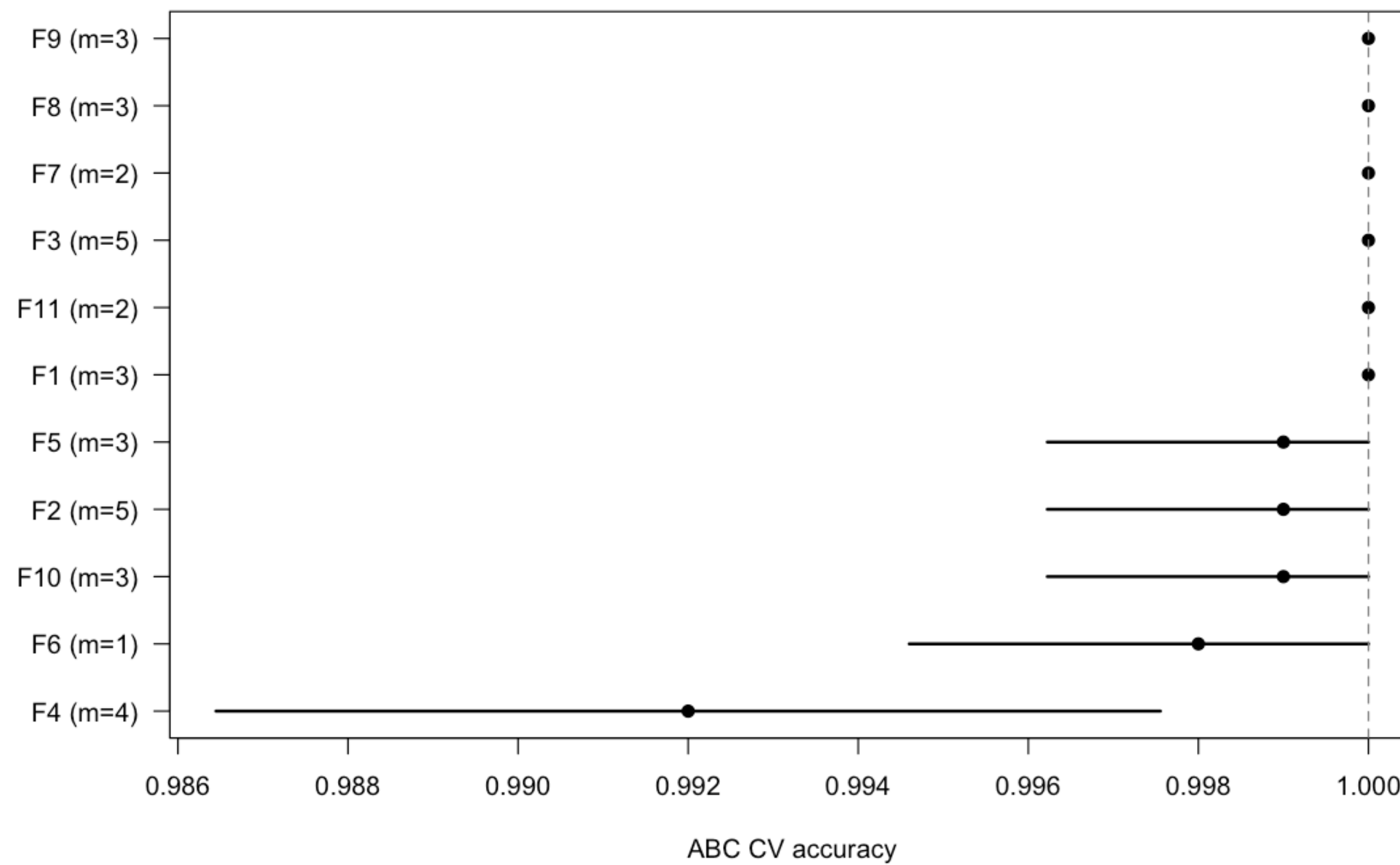
- one query (child of a missing person in my setup)
- m typed individuals (genotyped relatives available)

To turn multi-person evidence into one fixed-length input:

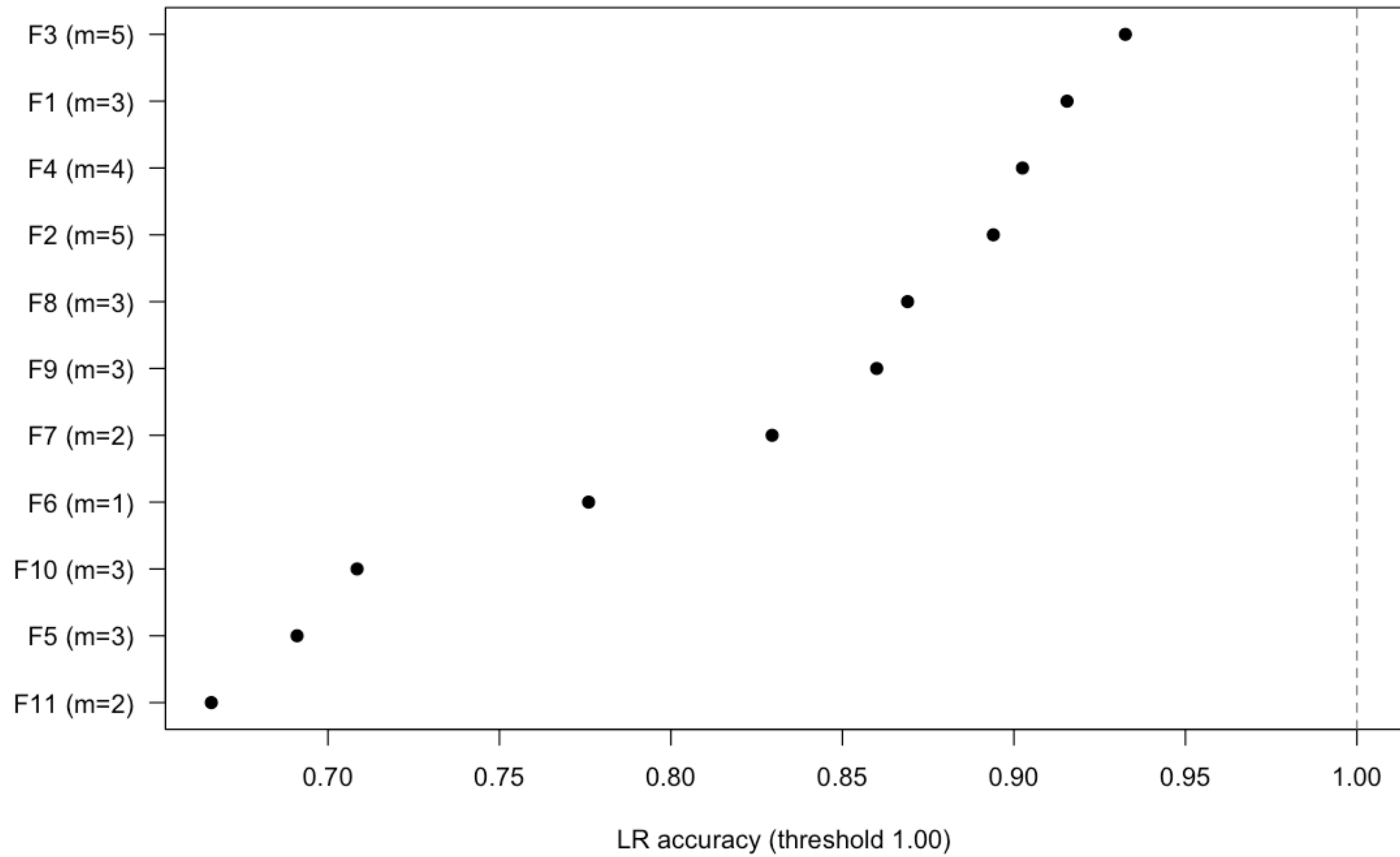
- I compute the 9 stats for each (query, typed_i) pair
- I stack them into slots, creating a single vector per replicate
- I use ranked slot order: slot 1 is the typed person sharing the most with the query, etc.

This is multi-person evidence via stacked pairwise summaries.

ABC CV accuracy by pedigree (mean $\pm$ 95% CI; cv_reps=5)



LR accuracy by pedigree (fixed threshold 1.00)



Limitations

- H2 does not preserve dependence among typed individuals when $m > 1$
- my “multi-person” is stacked pairwise evidence, not explicit triangulation summaries